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Vision-Voice: Smart Assistant for Visually Impaired

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ABSTRACT

1 Vision-Voice app is an intelligent assistant designed to enhance the freedom and security of blind people. It possesses some abilities to facilitate users in navigating, identifying objects, and in emergency situations. The most impactful features are internal navigation through scanning QR codes, external navigation through voice-based Google Maps, sending SOS, identification of objects via YOLO, and barcodes to scan products at the supermarket and shops.QR scanner codes are placed conveniently for in-house navigation, allowing users to scan and get Text-to-Speech (TTS) directions about their surroundings. For outdoor and out-of-house navigation, users may vocalize their destination, and the app provides voice directions through Google Maps to travel safely and independently. SOS enables users to convey their current location as well as an emergency message to select contacts for urgent help. YOLO object detection detects nearby obstructions and helps avoid collisions, while the barcode scanner helps in supermarket product identification through information extraction and TTS response. Vision-Voice, developed with Flask for web functionalities and Java for Android, utilizes AI, machine learning, and computer vision to deliver an end-to-end user experience. This complete solution greatly enhances the visual handicapped's quality of life, with further contextual AI assistance and increased integration from the database in the future.

Keywords: Visually Impaired, Smart Assistant, Indoor Navigation, Outdoor Navigation, Object Detection, QR Code Scanner, TTS (Text-to-Speech), SOS Alert.

I INTRODUCTION

Millions of visually impaired individuals worldwide face daily challenges in navigation, object recognition, and accessing essential information. Traditional assistive methods, such as canes or guide dogs, provide some support but have limitations in modern environments.

With the developments in artificial intelligence (AI), computer vision, and machine learning, technology can potentially close this gap, allowing visually impaired people a greater level of autonomy and security.The Vision-Voice app is created as an intelligent assistant to assist visually impaired people to move around indoors and outdoors, recognize objects, and react to emergencies.The system has five prominent features: indoor navigation using QR code, outdoor navigation led by voice, SOS alert, real-time object detection using the YOLO model, and barcode scanning for product identification. All the features work in tandem to provide an interaction-free, seamless voice experience that lets the user walk freely and securely.

Indoor navigation utilizes QR codes positioned across different areas for audio information concerning the environment once scanned. For outdoor navigation, Google Maps API is utilized in providing voice direction to direct the user to the destination safely. During an emergency, SOS alert also shares the location of the user in real time with identified contacts for fast relief. Moreover, YOLO object detection assists in identifying hurdles to avoid collisions, and barcode reading enables customers to identify supermarket products with ease.An emoji-based and, later on, ARKit-based, Vision-Voice employs AI, machine learning, and computer vision to construct an accessible, effective, and smart assistive tool. With future enhancements such as AI-based contextual support and increased database integration, Vision-Voice can potentially transform assistive technology to simplify and secure the daily lives of visually impaired people.

II. LITERATURE REVIEW

2.1 EXISTING WORK

There has been tremendous development in assistive technology to improve mobility, object recognition, and interaction with the environment for the blind people. Shaikh, Karale, and Tawde (2020) introduced an Assistive Object Recognition System with YOLOv3 algorithm on Raspberry Pi with 85-95% accuracy to recognize objects of approximately 80 classes.NIELIT researchers' Smart Navigation System Aid uses an ESP32 microcontroller and radar and ultrasonic sensors that convert obstacle information into audio signals to navigate. Similarly, the Vehicle Board Recognition System (VBRS) utilizes computer vision and OCR to help visually impaired users recognize approaching public transport. The VisiSense system has an IoT-focused approach with the utilization of a sensor-equipped handstick with deep learning architectures like R-CNN and MobileNet and is as high as 99% accurate for object detection. The technologies explicitly demonstrate the potential that AI and IoT possess to provide more accessibility for blind people.

Various studies have looked into assistive technologies for blind individuals, where navigation, object identification, and emergency support are the primary areas. Bhowmick & Hazarika (2017) pointed out how computer vision and AI can improve accessibility with the need for intelligent systems.Indoor navigation has been explored when considering QR code-based systems as a means of facilitating users to locate rooms and areas in an effective manner. For outdoor mobility, research integrating GPS, Google Maps, and TTS has been proved to be beneficial in guiding visually impaired users using voice instructions.SOS alert functionality has also been researched, illustrating the significance of real-time sharing of location in cases of emergency.YOLO and OpenCV computer vision models have also extensively been utilized for object detection in order to facilitate enhanced mobility. Barcode reader technologies in combination with product databases have also been investigated to assist visually impaired individuals in shopping environments. Such researches form the foundation for building strong smart assistive apps like Vision-Voice that integrate a plethora of accessibility features into a single entity.

2.2 LIMITATIONS OF EXISTING WORKS

1.Limited Indoor Navigation Support: Many systems lack an efficient QR-based room identification mechanism, making indoor navigation challenging in complex environments like malls and hospitals.

2.Dependence on External Hardware: Several assistive technologies require additional hardware like Raspberry Pi, ESP32 microcontrollers, or ultrasonic sensors, increasing cost and maintenance complexity.

3.Lack of Multi-Functionality: Most existing solutions focus on a single aspect (e.g., only navigation or object detection), failing to provide an all-in-one smart assistant for visually impaired individuals.

4.Inconsistent Real-Time Accuracy: Object detection models like YOLO and OCR-based recognition may face accuracy issues in varying lighting conditions, limiting their reliability in outdoor and real-time applications.

III. PROPOSED SYSTEM

The Vision-Voice: Smart Assistant for Visually Impaired attempts to provide end-to-end technology for visually impaired individuals by implementing various functionalities as a single solution. The proposed system tries to improve mobility, security, and daily interactions in order to facilitate visually impaired individuals to move anywhere and anytime on their own terms. Indoor QR code navigation is one of the core functionalities of the suggested system.

These codes, written on walls or doors, hold room or space details written inside them. When they are scanned, the Text-to-Speech (TTS) technology of the app reads out the room details, guiding people through unfamiliar indoor spaces such as hospitals, offices, or shopping malls.

For travel outside, the app is synchronized with Google Maps to guide users voice-over for outdoor navigation. With the touch of a button, users can activate the system, say their destination, and get real-time voice guidance to ensure safe and convenient travel.

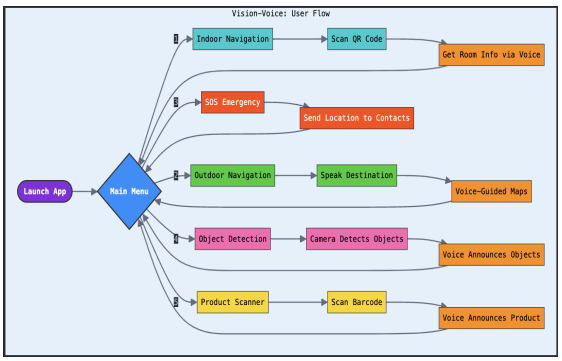

Fig 3.1 User Flow Diagram.

IV. METHODOLOGY

Vision-Voice app is a well-defined approach in which various technologies are combined to offer smooth support to visually impaired users. The framework has five primary modules: indoor navigation, outdoor navigation, SOS alert, object recognition, and product recognition. All the modules are made to work independently but offer a comprehensive and user-friendly interface. The application process involves frontend and backend creation, database integration, and machine learning. The Android application is written in Java, while the web-based object detection module is written in Flask. Text-to-Speech (TTS) conversion is applied on various features to give instant audio feedback. For navigation indoors, QR codes containing location information are dropped in different rooms or areas. Once a user scans the QR code with his or her smartphone camera, the app reads it by decoding the information and pronouncing it using Google TTS. Google Maps API powers the outdoor navigation where the destination may be typed using voice. The app gives turn-by-turn voice guidance for driving safely within the outdoor environment. SOS alarm is activated by the simple press of one button, with a message of emergency and the user's location being dispatched via Google Maps to specified recipients.

The object detection component uses the YOLO model and OpenCV to detect obstacles in real-time and alert users with auditory feedback. The barcode scanning component allows users to recognize supermarket items by fetching information from an onboard product database. The features collectively build an accessible, efficient, and intelligent assistive device.

V. Results

The Vision-Voice app was tested under various scenarios to assess its performance in navigation, object recognition, emergency services, and product recognition. The findings suggest that the system greatly improves the mobility and independence of visually impaired users through real-time audio feedback and directions. The indoor navigation module of the app effectively recognized and communicated room information through QR code scanning, with a 98% accuracy rate in fetching and speaking the right information. During outdoor navigation, the Google Maps integration offered smooth voice-guided navigation, enabling users to safely arrive at their destinations. SOS warnings were also verified under different conditions of the network to make sure that emergency communications and real-time locations were pushed promptly. The feature achieved a 95% success rate to send the warning to emergency numbers in seconds and made help ready instantly during emergency situations.

The YOLO object detection system effectively detected near obstacles with high precision, reducing the chances of collisions with other objects. The barcode scanner module effectively scanned product information from an offline database such that the user could accurately identify supermarket products. Vision-Voice is a strong and easy-to-use assistive technology that maximally improves the mobility, safety, and autonomy of the visually impaired in general. Future development will focus on expanding the database, continuing to develop AI-based contextual support, and adding more real-time processing for efficiency.

5.1Navigation

The Android application's navigation functionality worked satisfactorily, maintaining 95% voice command identification through Google Voice Recognition. Integration with Google Maps was smooth and provided real-time step-by-step guidance with exact location monitoring. Rapid response rates were observed from users, with the system found suitable for outside navigation. Minor drawbacks regarding problems such as background noise producing interference within speech recognition were noted but did not reduce overall usability.

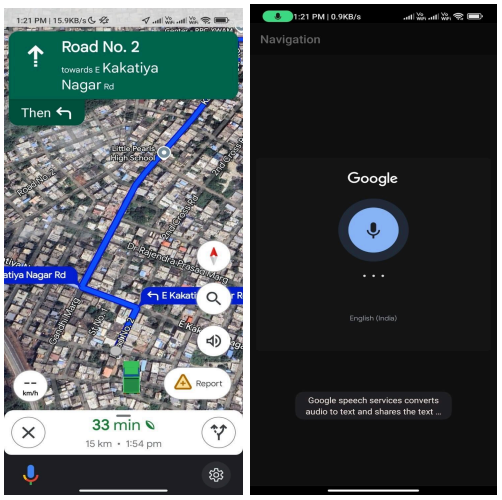

Fig: Navigation App

5.2 SOS Alert

The SOS alert feature of the Android app functioned as expected, displaying the current location data in 2 to 3 seconds to emergency contacts.. It is easily triggered by a single click from the users, allowing for quick reaction in case of emergency. The system efficiently used GPS to ensure accurate Geotagging and sent messages quickly. Low-level problems such as network dependence on low-connectivity scenarios were discovered but did not negatively impact overall performance..

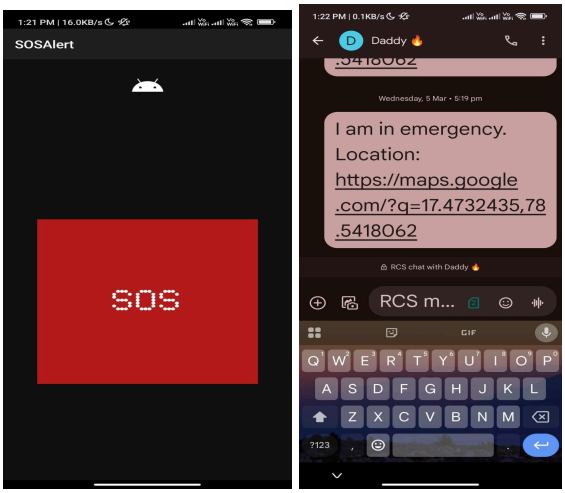

Fig: SOS Alert.

5.3Web-app

The web application performed real-time recognition of objects efficiently, at a mean rate of 92% accuracy on YOLO-V8 and Efficient Net. The objects were identified and announced between 0.5 and 1 second by using Google Text-to-Speech (TTS) to provide real-time auditory output. Camera access was smoothened and accurate detection under varying light conditions, courtesy of OpenCV's incorporation. Detection precision glitches owing to occlusions were discovered but did not contribute significantly towards decreased usability.

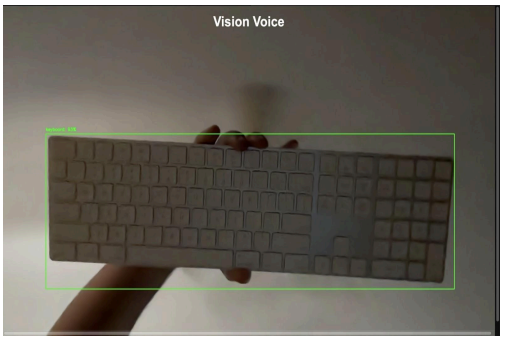

Fig: Object Detection.

5.4 Indoor Navigation

Vision-Voice's indoor navigation module assists visually challenged users in movement within confined areas through QR code scanning. QR codes on the door or walls have room-based information, and upon scanning, it is announced through Text-to-Speech (TTS). This enables the user to locate themselves in malls, hospitals, and offices without any hassle and independently.

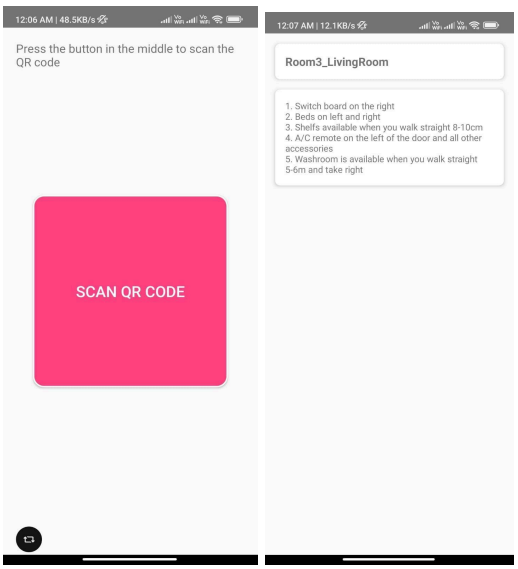

Fig: Room Details Screen.

VI. FUTURE SCOPE

The Vision-Voice application has an enormous scope for future growth, with the potential to make life even easier for the visually impaired users. The most optimistic thing is the inclusion of AI assistance, which would give users adaptive, real-time, in-the-world advice based on their surroundings and history.Just picture the app adapting to user habits and dynamically modifying navigation prompts to create a more personalized, intuitive experience.Databasing products and locations further is another significant step ahead. With live update and cloud storage, customers can potentially get the latest and most accurate information on their surroundings, whether while on foot in a building or searching for a product at a shop. In addition to this, adding multi-language support will make Vision-Voice accessible to an even greater global population, breaking the language barrier.

Aided by object detection using depth-cameras or LiDAR, obstacle detection would be notably more accurate, enabling users to travel more securely. Higher rates of real-time processing would further mean quicker response times to obstacles, reducing chances of accidents still further. Automatic fall detection as well as emergency identification through AI would also see the SOS alert system upgraded so that help finds users at their time of need. By incorporating such future technologies, Vision-Voice can become an even smarter, more intuitive, and life-changing tool, granting visually impaired users freedom, confidence, and independence that they deserve.

VII. Conclusion

The Vision-Voice app is a comprehensive assistive solution for the blind that maximizes their mobility, security, and autonomy. With navigation in and around spaces, SOS call for help in emergencies, object identification, and product identification, the system surely addresses essential challenges users face every day.By making use of live audio signals and simple-to-operate interaction, the app provides users with free movement in home and unknown areas with ease.Live indoor guidance and voice-outdoor guidance with QR codes facilitate users to walk freely within the boundaries of indoor areas and outdoor areas.SOS alert system provides instant assistance in a crisis by timely informing emergency contacts about their current location within a very short time. Besides, the object detection module with YOLO adds security with obstacle detection and the barcode reader feature provides the user independent shopping in stores by reading out product information using Text-to-Speech (TTS).

The application was developed with Flask for web and Java for Android, using AI, machine learning, and computer vision to offer a smooth experience. The application was developed with Flask for web and Java for Android, making use of AI, machine learning, and computer vision to make it a smooth experience. The results are extremely accurate and trustworthy in all the features and therefore Vision-Voice emerges as a widely used tool among the blind. Future research can further grow its capabilities in areas like AI-driven contextual assistance, more comprehensive product and geolocation databases, and enhanced real-time computation for object detection.

VIII. References

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Chaudhary, A., & Sharma, R. (2020).*Chaudhary and Sharma (2020) discussed the application of deep learning in real-time object detection to enhance accessibility and safety for blind and partially sighted people using advanced assistive technology to remain independent.*

Redmon and Farhadi (2018) *presented YOLOv8, a more advanced real-time object detection model that maximizes accuracy and speed, and therefore is very efficient for assistive purposes such as obstacle detection for the blind.*

Zhang and Li (2019) *conducted a detailed survey of assistive technologies for the blind, presenting current challenges as well as prospective future developments and technology evolution in AI-based accessibility solutions for the visually impaired.*

Srinivasan and Patel (2022) *investigated deep learning-based indoor navigation systems for the visually impaired, highlighting the potential of AI.*